SOUTH AUSTRALIA.

REPORT

OF THE

GOVERNMENT ASTRONOMER

TO THE

BOARD OF VISITORS

OF THE

ADELAIDE OBSERVATORY,

Read at the Annual Visitation of the Adelaide Observatory, February 16th, 1922.

UNIVERSITY OF VICTORIA

Adelaide:
BY AUTHORITY: R. E. E. ROGERS, GOVERNMENT PRINTER, NORTH TERRACE.

1924.

Digitized by Google

The following is a statement of meridian observations made during the period January 1st to December 31st, 1921 :—

Stars.	Observations in—	
	Right Ascension.	Declination.
Zone stars	221	222
Clock Stars for time and longitude	1,773	—
Total	1,994	222
Determinations of Collimation	142	—
" Level	205	—
" Nadir Point	22	—
" Azimuth	193	—

Azimuths were determined by combining stars North and South of the Zenith, results for clock error and azimuth being obtained by a graphic method, or sometimes from a least square solution.

The following table gives the mean collimation, level, and azimuth corrections for the transit circle telescope for each month during the same period. (The sign + in level means west end high, and in azimuth west end displaced northwards) :—

Month.	Collimation.	Level.	Azimuth.
1921.			
January	— ·07	+ ·36	— 1·46
February	+ ·03	— ·34	— 1·14
March	— ·13	— ·36	— 1·23
April	— ·08	— 1·18	— 1·06
May	— ·11	— 2·09	— ·87
June	— ·10	— 1·55	— 1·41
July	— ·12	— 1·08	— 1·20
August	— ·13	— ·59	— 1·89
September	— ·09	— ·29	— 1·42
October	— ·13	— ·01	— 1·82
November	— ·15	— 35	— 1·42
December	— ·09	— ·97	— 1·04

A further determination of the pivot errors of the transit circle telescope under the conditions of relief obtaining during the boundary determination was made in November and December. The results are consistent, and show errrors of appreciable magnitude. The necessary corrections are being applied to the Adelaide observations.

The Cooke Equatorial.

This instrument, of 8in. aperture, has been used principally in connection with public visits, for which Friday evenings have been set aside, as in previous years, and also for occasional and miscellaneous observations. Mr. O. Glastonbury, of the South Australian Astronomical Society, continued his observations of the variable stars R. Centauri and S. Carinæ with this telescope, in accordance with suggestions made by Professor H. H. Turner, of Oxford.

III.—FIELD WORK.

A—Interstate and Commonwealth Boundary (129th Meridian).

In my last report to the Board of Visitors (April 4th, 1921), an account was given of the preparations and experiments made for determining the 129th meridian by the aid of long distance signals from the French high power wireless stations at Lyons, enabling us to make a direct measurement of longitude from Greenwich to Australia.

A trial of the method was made at the boundary station at Deakin, on the Trans-Australian Railway, in November, 1920, with highly successful results, described in my report.

The final work of determining the boundary was therefore decided upon.

The success obtained had shown that the second and simpler programme, which entailed a much reduced expenditure, could be carried out.

A meeting of the Longitude Committee was held in Melbourne in February, 1921, and it was decided that we should aim at determining directly, by long distance wireless signals, two points on the 129th meridian, viz., Deakin in the south and Kimberley in the north, and that such observatories as could do so should simultaneously receive the special signals, so as to redetermine their own longitude direct from Greenwich, and incidentally strengthen the boundary determination by providing a check as well as an extra longitude link.

The Minister for Home and Territories then agreed (February 4th, 1921) on behalf of the Commonwealth Government to contribute £500 towards the undertaking.

The party arrived at Deakin on April 15th. Mr. C. A. Maddern went with me in advance, and we made a series of magnetic observations at Cook on April 13th and 14th, and afterwards a similar series at Deakin, in addition to the longitude observations. Mr. H. B. Curlewis, Government Astronomer of Western Australia, met us at Deakin ; and the other members of the party were Messrs. C. M. Hambidge and A. Williams, of the Survey Department ; and H. S. Duncan, who came as a volunteer assistant, and gave us valuable help.

The

Digitized by Google

The Defence Department again lent us the portable wireless mast and aerial, but Mr. Bowen's services could not be spared for the final work.

As on the occasion of the preliminary observations, the Commonwealth Railway officials did everything they could to help the party, and the assistance of the gangers in establishing and dismantling the camp, the use of the station cabin as wireless room, and the loan of a set of accumulator cells for the wireless amplifier, were much appreciated.

General Ferrié again very kindly arranged a series of wireless signals from Lyons at 20h. 15m. Greenwich Civil Time (corresponding to 5 a.m. " Central " or Railway Time at the Border), and in addition sent us a cable announcing that the Lafayette station at Bordeaux, at this time the most powerful in the world, would send out signals at 20h. G.C.T. (4·45 a.m. Central Time at the Border), on triple power and wave length 23,400 metres.

Admiral Hoogewerff also most kindly arranged to send out a special series of signals from Annapolis, near Washington, U.S.A., at 19h. 55m. G.C.T.

All these signals were sent at the above times each day from April 21st to May 10th, 1921.

We received them at Deakin as follows :—

Bordeaux, 20h., G.C.T., April 21, 22, 23, 24, 26, 27, 28, 30, May 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Total 18 times.

Lyons, 20h. 15m., G.C.T., April 22, 23, 24, 25, 26, 27, 28, 29, 30, May 1, 2, 3, 4, 5, 7, 9, 10. Total 17 times.

Annapolis, 19h. 55m., G.C.T. (special series), April 26, 27, 29, 30, May 3, 5, 6, 7, 8, 9, 10. Total 11 times.

Annapolis, 3h., G.C.T. (regular signals), April 25, 26, 27, 28, 29, 30, May 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11. Total 17 times.

In addition to the reception and recording of these signals at Greenwich, Paris, and Washington, they were received at Ottawa (Bordeaux and Annapolis signals only), Helwan, near Cairo (Lyons signals), Sydney (Lyons and Bordeaux signals), Perth (Lyons and Bordeaux signals), and Adelaide (Lyons and Bordeaux signals).

It is worthy of note that we were able to receive in the field the special signals from all the transmitting stations, and that this is the first longitude determination to be made with the help of wireless time signals completely girdling the earth.

In addition to the signals from Lyons, Bordeaux, and Annapolis, we also took night and morning signals from Adelaide and from Perth, as an aid in checking our chronometer rate and getting local time.

For the determination of local time star observations were taken by Mr. Curlewis with the Perth almucantar, and by myself with the portable transit instrument lent by the Sydney Lands Department and brought up to date in Adelaide. Mr. Curlewis also used the almucantar instrument, and Mr. Hambidge his theodolite, for latitude determination.

The following are the preliminary results, uncorrected as yet for personal equation and instrumental errors, which are to be found by comparison work at Greenwich :—

LONGITUDE OF DEAKIN, FROM WIRELESS SIGNALS FROM LYONS, BORDEAUX, AND ANNAPOLIS, APRIL-MAY, 1921.

Provisional Results.

European Date.	Lyons Signals Received at—			
	Greenwich and Deakin.	Paris and Deakin.	Sydney and Deakin.	Adelaide and Deakin.
1921.	h. m. s.	h. m. s.	h. m. s.	h. m. s.
April 21	—	—	—	—
22	8 35 52·49	8 35 52·71	8 35 52·90	—
23	52·39	52·60	52·81	—
24	52·44	52·68	52·86	8 35 52.81
25	52·50	52·70	52·89	52.84
26	52·49	52·69	52·94	52.77
27	52·56	52·74	52·92	52.88
28	52·44	52·63	(52·76)	52.80
29	52·57	52·77	53·01	52.82
30	52·52	52·75	52·97	—
May 1	52·63	52·80	53·09	52.82
2	52·48	52·64	52·88	52.82
3	52·54	52·70	52·94	52.84
4	52·49	52·68	—	52.82
5	52·53	52·64	—	52.84
6	—	—	—	—
7	52·59	52·65	53·09	52.91
8	—	—	—	—
9	52·56	52·61	—	—
10	52·68	52·75	—	52.90
	8 35 52·52	8 35 52·69	8 35 52·93	8 35 52.84
Transmission time correction	−·04	−·04	—	—
	8 35 52·48	8 35 52·65	8 35 52·93	8 35 52·84

Digitized by Google

LONGITUDE OF DEAKIN, &c.—*continued.*

Provisional Results.

European Date.	Bordeaux Signals Received at— Ottawa and Deakin.	Paris and Deakin.	Adelaide and Deakin.	Adelaide Signals at Adelaide and Deakin.
1921.	h. m. s.	h. m. s.	h. m. s.	h. m. s.
April 21	8 35 52·73	8 35 52·87	—	—
22	52·58	52·71	8 35 52·74	8 35 52·79
23	52·55	52·82	—	—
24	52·57	52·67	52·76	52·81
25	—	—	—	52·80
26	—	52·70	52·82	—
27	—	52·75	52·87	52·83
28	52·54	—	52·71	52·77
29	—	—	—	—
30	52·71	52·78	—	52·72
May 1	52·72	52·74	52·76	52·84
2	52·79	52·80	52·91	52·84
3	52·65	52·68	52·83	52·76
4	52·59	52·60	52·76	52·70
5	52·65	52·60	52·87	52·71
6	52·72	52·69	52·90	52·77
7	52·67	52·64	52·87	52·90
8	52·68	52·61	52·89	—
9	52·63	52·83	52·86	52·86
10	52·79	52·73	—	—
	8 35 52·66	8 35 52·72	8 35 52·82	8 35 52·79
Transmission time correction	—·03	—·04	—	—
	8 35 52·63	8 35 52·68	8 35 52·82	8 35 52·79

European Date.	Annapolis Signals Received at— Ottawa and Deakin.	Washington and Deakin.
1921.	h. m. s.	h. m. s.
April 21	—	—
22	—	—
23	—	—
24	—	—
25	—	—
26	8 35 52·66	8 35 52·68
27	52·63	52·66
28	—	—
29	52·75	52·77
30	52·63	52·68
May 1	—	—
2	—	—
3	52·65	52·67
4	—	—
5	52·64	52·70
6	52·67	52·74
7	52·74	52·75
8	52·77	52·81
9	52·65	52·68
10	52·75	52·78
	8 35 52·69	8 35 52·72
Transmission time correction	—·05	—·05
	8 35 52·64	8 35 52·67

The above results are not final; corrections for personal and instrumental errors are to be applied. The position of the 129th meridian in accordance with these figures is approximately $1\frac{3}{4}$ miles east of the observing site, marked by the concrete blocks on which the portable transit and almucantar instruments were mounted.

The Surveyor-General of South Australia (Mr. T. E. Day) visited our camp during the course of the work. We much appreciated his assistance throughout, as also that of his predecessor in office, Mr. N. W. Pethick. They were naturally anxious to have the initial points of the survey determined with the greatest possible accuracy.

KIMBERLEY.

When the work at Deakin was almost finished we received from Mr. J. T. H. Goodwin (Commonwealth Surveyor-General and representative on the Longitude Committee) the following telegram, dated April 29th, 1921 :—" Your letter 19th instant. Minister desires to inform you he has no knowledge any arrangements with Durack. Cabinet have now agreed to contribute £500 on understanding it is the limit of Commonwealth obligation directly or indirectly. The States concerned are to enter into an agreement with the Commonwealth that the two points determined shall for ever be recognised as on the boundary, and the line joining these two points shall be the boundary. Minister says longitude party must make its own transport arrangements."

The

Digitized by Google

The Premier of South Australia received, on May 4th, a similar telegram from the Acting Prime Minister.

Messrs. Curlewis and Hambidge and I went on to Perth *en route* for Kimberley. Mr. Maddern had returned to the Adelaide Observatory. At Perth we worked at some of the Deakin results during the interval before the departure of the steamer, and I took the opportunity of making the necessary personal equation obervations with the Perth Transit Observer (Mr. Nossiter). In response to my telegram the Hon. the Minister of Education instructed Mr. Maddern to rejoin and go on with the expedition.

We left Fremantle by the s.s. *Bambra* on May 28th, and arrived at Wyndham on June 12th. Mr M. P. Durack, M.P. for Kimberley, met us at Wyndham, and delayed his return to Perth in order to help us. We appreciated very greatly his support and practical help, and the hospitality so kindly extended to us by himself and his cousin at their stations of Ivanhoe and Argyle; also their help with the transport of the party and equipment by motor car and buggy, which saved us considerable inconvenience and expenditure, and we cannot speak too highly of their kindness.

We established our camp on June 19th near the reputed position of the boundary, 18 miles N.E. of Argyle Station. The track was partly over unbroken ground, and was made for us by a team of Mr. Durack's mules. Unfortunately one was killed owing to a stampede on the way. For our benefit Mr. Durack rode out to a prominent mountain, and re-established the trig. survey mark on its summit. Mr. Hambidge made a triangulation connection with this and other points. This one we named Mount Ferrié (and it is so entered in the Western Australian survey maps), in honor of the Director of the French radio service, who enabled us to do this work by the wireless time signals. Other mountains were named after Sir Frank Dyson (Astronomer Royal of England), Admiral Hoogewerff, and members of our party, including Mr. Durack.

On June 20th the wireless masts and aerial were erected and instruments were set up and adjusted. The same evening we received the Adelaide wireless longitude signals. On the following morning we heard the first of the Kimberley series of signals from Lyons. Reception conditions were not so good as at Deakin, on the Nullarbor Plain, atmospheric absorption of the wireless waves being greater, and we could not get the Annapolis signals at all in the early morning, but fortunately received their mid-day ones on several occasions. For some obscure reason which we could not discover the wireless receiver did not function well here on the very long wave lengths used by Bordeaux (23,400 metres), so that we only heard these signals on a few occasions. Sydney Observatory experienced a similar difficulty, but Adelaide and Perth received Bordeaux quite well, although they failed to hear Annapolis.

The following are the dates on which signals were received at Kimberley :—

Bordeaux, 20h., G.C.T., June 23, 26, 27, 28, July 1. Total, 5 times.

Lyons, 20h. 15m., G.C.T., June 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, July 1, 2, 3, 5, 6, 7. Total, 17 times.

Annapolis, 19h. 55m., G.C.T., Nil.

Annapolis, 3h., G.C.T., June 23, 24, 25, 26, 27, 28, 29, 30, July 1, 2, 3, 5, 7, 8. Total, 14 times.

As at Deakin, night and morning signals from Adelaide and Perth were received to check chronometer rate and assist in the determination of local time. Up to the present we have not received results of reception at Greenwich and Paris Observatories. The Dominion Observatory of Canada, being engaged in special wireless longitude work in connection with the Mackenzie River Oil Fields, could only give a few results from the Bordeaux signals. The Director (Dr. Klotz) completed the last great chain of longitude round the earth by cable, in his work across the Pacific in 1903-4, and has taken a very great interest in this work of ours.

The following preliminary results for the longitude of our Kimberley station are based on the results of reception of Annapolis and Adelaide signals, a list of corrections for Annapolis having been supplied by Admiral Hoogewerff :—

KIMBERLEY LONGITUDE.

Provisional Values.

From Adelaide Comparison.

1921.	h.	m.	s.
June 23 9h	8	36	05·67
24			·81
25		—	
26		—	
27			·75
28		—	
29			·84
30			·80
July 1			·75
2			·80
3			·74
4		—	
5			·83
6			·85
7			·85
Mean	8	36	05·79

From Annapolis Comparison.

1921.	h.	m.	s.
June 23 0h	8	36	05·63
24			·49
25			·57
26			·64
27			·64
28			·59
29			·60
30			·53
July 1			·59
2			·59
3			·62
4		—	
5			·47
6		—	
7			·60
8			·54
Mean	8	36	05·58
Transmission time			—·05
Longitude	8	36	05·53

The difference between these two results suggests that the adopted Adelaide longitude is somewhat too high, but the results from other stations are needed before a full discussion of the value for Kimberley can be determined, and, as in the case of Deakin, it is necessary for personal equation and instrumental errors

errors to be eliminated before finality is reached. The preliminary result obtained shows that the 129th meridian in the Kimberley district is approximately between 1¼ and 1½ miles west of the observing site. The latter is permanently marked by the two concrete pedestals on which the portable transit and almucantar instruments were fixed. The approximate distance to the 129th meridian was chained off by Mr. Hambidge and temporarily marked.

We were visited by parties from Argyle homestead, including some of the natives employed there. It was strange to see these people of the stone age in contact with the modern invention of wireless telegraphy hearing signals from the other side of the world. On our return to Argyle they gave a special corroboree in honor of the work.

Fortunately all circumstances favored the Kimberley expedition throughout, and we were able, through Mr. Durack's help, to reach Wyndham just in time to catch the *Bambra* on her next return journey to Fremantle. As she was not coming back for another three months, this meant a saving of considerable expense.

There still remains the important problem of eliminating the errors due to differences between the Greenwich and Australian instruments and observers. It is always desirable to do the comparison work within as short an interval of time as possible, so that little change will have taken place in these. I considered, moreover, how much time and expense would be saved if I could immediately proceed to Greenwich *via* Singapore and Colombo, and communicated with the Hon. Minister of Education by telegraph about this, which has always been part of the projected programme (the original programme, indeed, having included visits by two Australian observers to Greenwich). I did not, however, receive Ministerial approval. The instrumental error and personal equation with the Greenwich observer and instrument have yet to be determined. This is a pressing matter, as it is holding up the completion of the boundary results.

B.—ADELAIDE OBSERVATORY LONGITUDE OBSERVATIONS.

While the longitude expedition was at Deakin and Kimberley the observations at Adelaide Observatory were in charge of Mr. J. B. O. Hosking, Chief Assistant, who carried out the important work here with great care and ability.

The signals emitted from this Observatory, through the medium of the Adelaide Radio Station, were of a high order of accuracy, and were controlled by the meridian observations with the transit circle telescope. The reliability of these observations and of the Adelaide signals, was a source of great satisfaction to us at the boundary. The signals from Lyons and Bordeaux were received at the Observatory by Messrs. E. A. Thrum and G. R. Fuller, to whom much credit is due for their enthusiasm and success in obtaining exact records of the long distance signals. They voluntarily spent the nights in the wireless room after taking stars, so as to be sure to be on hand in good time for the early morning work.

Mr. Thrum devised a specially valuable method for receiving the signals, his account of which I am forwarding to the Royal Astronomical Society. The results of the reception at Adelaide will be given later.

I acknowledge with sincere thanks the great help afforded to me in carrying out the important work of the boundary determination by the support of the Board of Visitors.

Mr. Curlewis and I are also especially indebted to the Director of the Commonwealth Radio Service, and to the officers in charge and staff of the Adelaide and Perth Radio Stations for the courtesy extended to us, and for the great assistance given in connection with the transmission of signals and loan of apparatus.

A full joint report of the undertaking will be made by Mr. Curlewis and myself when the observations are fully worked out, and after investigation of personal equation, &c., at Greenwich. The cost of printing the report will be part of the Western Australian contribution.

SUMMARY OF FIELD WORK CARRIED OUT BY THE ADELAIDE OBSERVATORY IN RECENT YEARS.

With the conclusion of the important work of determining the 129th meridian in Australia, it is fitting that I should present a summary of all the field operations carried out in recent years by the Adelaide Observatory. These comprise (1) latitude and longitude, (2) magnetic observations in three different periods and divisions of the State.

I. Eastern boundary (the 141st meridian).

II. Points in the geodetic and magnetic survey of South Australia from Port Augusta to Mount Gambier, and from Oodnadatta to near the Musgrave Ranges.

III. The 129th meridian.

These are given in the accompanying table.

I deeply regret that during my absence on the boundary expedition the usual line for latitude and longitude work (£150) was deleted from the Estimates. I trust that it will be re-established on the next Estimates.

FIELD

Digitized by Google

FIELD WORK CARRIED OUT BY OBSERVATORY DEPARTMENT—*continued.*

Date.	Station.	Nature of Observations.	Observers.	Total Amount Provided (but not always Spent).
1917.	II.—Geodetic and Magnetic Survey of South Australia—*continued.*			
May ...	Blanchetown	Latitude, longitude, and magnetic	G. F. Dodwell and Prof. Kerr Grant	—
Nov. ...	Port Pirie	Magnetic.....	G. F. Dodwell and Prof. Kerr Grant	1917-18, latitude and longitude, £150
1918.				
Jan. ...	Nairne.....	Latitude, longitude, and magnetic	G. F. Dodwell and Prof. Kerr Grant	Magnetic survey, £150
Jan. ...	Goolwa	Latitude, longitude, and magnetic	G. F. Dodwell and Prof. Kerr Grant	—
Feb. ...	Pt. Macdonnell	Latitude, longitude, and magnetic	G. F. Dodwell and Prof. Kerr Grant	—
Feb. ...	Mt. Ruskin	Latitude, longitude, and magnetic	G. F. Dodwell and Prof. Kerr Grant	—
March ..	Paradise	Magnetic.....	G. F. Dodwell, Prof. Kerr Grant, and W. R. Blackman	—
April ...	Althorpe Island	Magnetic	G. F. Dodwell	—
April ...	Wincoby Island	Magnetic	G. F. Dodwell	—
April ...	Spencer Gulf	Magnetic observations on s.s. *Conqueror*	G. F. Dodwell and J. I. McLauchlan	1918-19, latitude and longitude, £150; magnetic survey, £150
1919—No field work during the financial year July, 1918-June, 1919; £128 from latitude and longitude vote and £108 from magnetic vote spent on instruments, computations, and printing.				
1920.				
Aug. ...	Ardrossan	Magnetic.....	G. F. Dodwell and C. A. Maddern	—
Aug. ...	Maitland.....	Magnetic		
	III.—Western Boundary.			
1920.				
April ...	Sydney Observatory...	Testing portable transit instrument and trial reception overseas wireless signals	G. F. Dodwell and W. E. Raymond	1919-20, latitude and longitude, £300; magnetic survey, £150
Nov. ...	Deakin, S.A. W.A. boundary	Latitude and longitude. Trial reception of Lyons signals for boundary observations	G. F. Dodwell, C. A. Maddern, C. M. Hambidge, and V. D. Bowen, in co-operation with Mr. Curlewis, Government Astronomer of W.A.	—
1921.				
Apl.-May	Cook	Magnetic.....	G. F. Dodwell and C. A. Maddern	—
Apl.-May	Deakin	Latitude, longitude, and magnetic. Final determination of boundary, southern point	G. F. Dodwell, C. A. Maddern, and C. M. Hambidge, in co-operation with Mr. Curlewis	—
May ...	Perth Observatory ..	Personal equation observations ..	G. F. Dodwell and C. Nossiter	—
Jun.-July	Kimberley, N.T./W.A. boundary	Latitude and longitude. Final determination, northern point on 129th meridian	G. F. Dodwell, C. A. Maddern, and C. M. Hambidge, in co-operation with Mr. Curlewis	—
				1920-21, latitude and longitude, £300; magnetic survey, £150

It will be seen that the Observatory has accomplished a great deal of field work in comparison with the small amount of expenditure, for during several years part of the amount voted for latitude and longitude work, and a larger part of the magnetic vote being unused lapsed to the Treasury. This would, however, have been needed for printing the results had the staff had time to work up all the computations, as we hope to do in the near future.

V.—INTERNATIONAL GEODETIC AND ASTRONOMICAL CONFERENCE.

This opens on April 20th, 1922, in Rome. Australia has been invited to join officially and send representatives. Mr. (now Professor) W. E. Cooke represented Australia at the Paris Conference, being sent by his Government, and made a very valuable contribution to the discussions, particularly in regard to his method of zone observations, which was adopted by the Conference, and at the same time gained information as to the international work being done on the world chart and catalogue of stars, in connection with which certain zones were allotted to Australia. The Adelaide Observatory takes a share in this work. Moreover ts geodetic work, and especially the recent determination of the 129th meridian, provide a good reason for Adelaide representation at the forthcoming conference at Rome. I have requested the Government to send me, and the Surveyor-General has supported my request.

VI.—TOTAL SOLAR ECLIPSE OF SEPTEMBER 21st, 1922.

A meeting of prominent citizens was held in the Adelaide Town Hall on November 28th, 1921. It was attended by His Excellency the Governor, and presided over by the Lord Mayor. Resolutions to send an expedition to Cordillo Downs, and to invite a British expedition to join it were carried. Strong general and executive committees (including the Board of Visitors of the Adelaide Observatory) were elected.

Digitized by Google